FLOX Therapy

INDICATIONS FOR USE:

INDICATION	ICD10	Regimen Code	HSE approved Reimbursement Status*
Adjuvant treatment of stage II or III colon cancer after complete resection of primary tumour	C18, C19, C20	00486a	N/A

*This applies to post 2012 indications

TREATMENT:

The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patients individual clinical circumstances.

- Folinic acid is administered weekly for 6 consecutive weeks (on days 1, 8, 15, 22, 29, and 36 of the treatment cycle), followed by a 2-week rest period.
- 5-Fluorouracil is administered as an intravenous bolus 1 hour after the folinic acid infusion has begun, and is administered weekly for 6 weeks (on days 1, 8, 15, 22, 29, and 36 of the treatment cycle), followed by a 2-week rest period.
- Oxaliplatin is administered as a 2-hour infusion before folinic acid and 5-Fluorouracil on days 1, 15, and 29 of the treatment cycle.

Treatment is administered for three 8-week cycles of therapy for a total treatment duration of 24 weeks (6 months) unless disease progression or unacceptable toxicity develops, whichever occurs first.

Facilities to treat anaphylaxis MUST be present when systemic anti-cancer therapy (SACT) is administered.

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Admin. Order	Day	Drug	Dose	Route	Diluent & Rate	Cycle
1	1, 15 and 29	Oxaliplatin	85mg/m^2	IV infusion	500mL Glucose 5% over 2 hours	Every 8 weeks for 3 cycles
2	1, 8, 15, 22, 29, 36	Folinic acid (Calcium leucovorin)	500mg/m^2	IV infusion	250mL Glucose 5% over 2 hours	Every 8 weeks for 3 cycles
3	1, 8, 15, 22, 29, 36	*5-Fluorouracil	500mg/m^2	IV bolus to be given 1 hour after start of the folinic acid infusion		Every 8 weeks for 3 cycles
Oxaliplatin is incompatible with 0.9% NaCl. Do not piggyback or flush lines with normal saline. For oxaliplatin doses ≤ 104mg use 250mL glucose 5%. Increase infusion rate time to 4 – 6 hours in case of laryngopharyngeal dysaesthesia reaction. Oxaliplatin administration must always precede the administration of 5-Fluorouracil. Oxaliplatin may be given at the same time as Folinic Acid *(Calcium Leucovorin)* using a Y connector. The 5-Fluorouracil bolus can then be administered immediately on completion of the folinic acid infusion.						
Folinic Acid *(Calcium Leucovorin)* must be administered prior to 5-Fluorouracil. It enhances the effects of 5-Fluorouracil by increasing 5-Fluorouracil binding to the target enzyme thymidylate synthetase. Acute neurotoxicity is common with oxaliplatin and can be precipitated on exposure to the cold therefore in this regimen patients should NOT suck on ice chips during the bolus injection of 5-Fluorouracil.						
*See dose modifications section for patients with identified partial dihydropyrimidine dehydrogenase (DPD) deficiency						

Note: Administration volumes and fluids have been standardised to facilitate electronic prescribing system builds.

ELIGIBILITY:

- Indications as above
- ECOG 0-2
- Adequate haematological, renal and liver function

CAUTION:

- Prior history of cardiac history
- Peripheral neuropathy with functional impairment prior to first cycle

EXCLUSIONS:

- Hypersensitivity to oxaliplatin or any of the excipients
- Breastfeeding
- Known complete dihydropyrimidine dehydrogenase (DPD) deficiency

PRESCRIPTIVE AUTHORITY:

The treatment plan must be initiated by a Consultant Medical Oncologist.

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TESTS:

Baseline tests:

- FBC, liver and renal profile
- ECG (if patient has compromised cardiac function)
- DPD testing prior to first treatment with 5-Fluorouracil using phenotype and/or genotype testing unless patient has been previously tested
    - In patients with moderate or severe renal impairment, blood uracil levels used for dihydropyrimidine dehydrogenase (DPD) phenotyping should be interpreted with caution, as impaired kidney function can lead to increased uracil blood levels. Consequently, there is an increased risk for incorrect diagnosis of DPD deficiency, which may result in under dosing of 5-Fluorouracil or other fluoropyrimidines, leading to reduced treatment efficacy. Genotype testing for DPD deficiency should be considered for patients with renal impairment.

Regular tests:

- FBC, liver and renal profile prior to each treatment
- ECG as clinically indicated
- Evaluate for peripheral neuropathy every 2 cycles

Disease monitoring:

Disease monitoring should be in line with the patient's treatment plan and any other test/s as directed by the supervising Consultant.

DOSE MODIFICATIONS:

- Any dose modification should be discussed with a Consultant
- Consider a reduced starting dose in patients with identified partial DPD deficiency
    - Initial dose reduction may impact the efficacy of treatment
    - In the absence of serious toxicity, subsequent doses may be increased with careful monitoring

Haematological:

Table 1: Dose modifications for haematological toxicity

ANC ($x10^9$/L)		Platelets ($x10^9$/L)	Dose
<1	and/or	<100*	Delay treatment until recovery
<0.5	and/or	<50	Delay treatment until recovery and consider reducing oxaliplatin and 5-Fluorouracil by 25% for subsequent cycles
Febrile Neutropenia			

*Given the slower accumulation of thrombocytopenia with oxaliplatin, the treating clinician may decide to proceed with treatment when platelets are 75 to 100 x 10^9/L

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Renal and Hepatic Impairment:

Table 2: Dose modifications in renal and hepatic impairment

<table>
<tr><th>Drug</th><th colspan="2">Renal impairment</th><th colspan="4">Hepatic impairment</th></tr>
<tr><td rowspan="4">Oxaliplatin[a]</td><td>CrCl (mL/min)</td><td>Dose</td><td colspan="4" rowspan="4">No dose adjustment is needed</td></tr>
<tr><td>>30</td><td>No dose adjustment is needed.</td></tr>
<tr><td><30</td><td>Consider 50% of the original dose</td></tr>
<tr><td>Haemodialysis</td><td>Consider 50% of the original dose, Haemodialysis within 1.5 hour after administration.</td></tr>
<tr><td rowspan="4">5-Fluorouracil[b]</td><td colspan="2" rowspan="4">No need for dose adjustment is expected.

Haemodialysis: No need for dose adjustment is expected.</td><td>Bilirubin (micromol/L)</td><td></td><td>AST</td><td>Dose</td></tr>
<tr><td><85</td><td></td><td><180</td><td>100%</td></tr>
<tr><td>>85</td><td>or</td><td>>180</td><td>Contraindicated</td></tr>
<tr><td colspan="4">Clinical decision.
Moderate hepatic impairment; reduce initial dose by 1/3.
Severe hepatic impairment, reduce initial dose by 1/2.
Increase dose if no toxicity.</td></tr>
<tr><td colspan="6">[a]Renal and hepatic dose modifications taken from Giraud et al 2023
[b]Renal recommendations from Giraud et al 2023, hepatic recommendations from NLCN</td></tr>
</table>

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Management of adverse events:

Table 3: Dose Modifications for Adverse Events

Adverse reactions	Recommended dose modification
*Peripheral neuropathy Grade 2 present at start of cycle Grade 3 or Grade 4	Reduce oxaliplatin by 25% if persistent, consider omitting oxaliplatin Discontinue oxaliplatin
Laryngo-pharyngeal dysaesthesia	Increase infusion time from 2 to 6 hrs
Mucositis and Stomatitis or Diarrhoea Grade 2 • 1st occurrence • 2nd occurrence • 3rd occurrence • 4th occurrence Grade 3 or 4 • 1st occurrence • 2nd occurrence	Delay treatment until toxicity has resolved to grade 1 or less and reduce dose for subsequent cycles as follows • No dose reduction • Reduce oxaliplatin and 5-Fluorouracil by 25% • Reduce oxaliplatin and 5-Fluorouracil by 50% • Withhold chemotherapy Delay treatment until toxicity has resolved to grade 1 or less and reduce dose for subsequent cycles as follows • Reduce oxaliplatin and 5-Fluorouracil by 50% • Withhold chemotherapy
Hand foot syndrome Grade 2 • 1st occurrence • 2nd occurrence • 3rd occurrence • 4th occurrence Grade 3 or 4 • 1st occurrence • 2nd occurrence	Delay treatment until toxicity has resolved to grade 1 or less and reduce dose for subsequent cycles as follows • No dose reduction • Reduce 5-Fluorouracil by 25% • Reduce 5-Fluorouracil by 50% • Omit 5-Fluorouracil Delay treatment until toxicity has resolved to grade 1 or less and reduce dose for subsequent cycles as follows • Reduce 5-Fluorouracil by 50% • Omit 5-Fluorouracil

SUPPORTIVE CARE:

EMETOGENIC POTENTIAL:

- As outlined in NCCP Classification Document for Systemic Anti Cancer Therapy (SACT) Induced Nausea and Vomiting- Available on the NCCP website

Oxaliplatin: Moderate (**Refer to local policy).**
5-Fluorouracil: Low (**Refer to local policy).**

For information:
Within NCIS regimens, antiemetics have been standardised by Medical Oncologists and Haemato-oncologists and information is available in the following documents:

- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Haemato-oncology) - Available on the NCCP website

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PREMEDICATIONS: Not usually required unless patient has had a previous hypersensitivity.

OTHER SUPPORTIVE CARE: Anti-diarrhoeal treatment (**Refer to local policy**)

ADVERSE EFFECTS:

- Please refer to the relevant Summary of Product Characteristics (SmPC) for details

REGIMEN SPECIFIC COMPLICATIONS

Oxaliplatin

- **Laryngopharyngeal dysaesthesia:** An acute syndrome of laryngopharyngeal dysaesthesia occurs in 1% - 2% of patients and is characterised by subjective sensations of dysphagia or dyspnoea/feeling of suffocation, without any objective evidence of respiratory distress (no cyanosis or hypoxia) or of laryngospasm or bronchospasm. Symptoms are often precipitated by exposure to cold. Although antihistamines and bronchodilators have been administered in such cases, the symptoms are rapidly reversible even in the absence of treatment. Prolongation of the infusion helps to reduce the incidence of this syndrome.

5-Fluorouracil:

Myocardial ischaemia and angina: Cardiotoxicity is a serious complication during treatment with 5-Fluorouracil. Patients, especially those with a prior history of cardiac disease or other risk factors, treated with 5-Fluorouracil, should be carefully monitored during therapy.

- **DPD deficiency:** DPD is an enzyme encoded by the DPYD gene which is responsible for the breakdown of fluoropyrimidines. Patients with DPD deficiency are therefore at increased risk of fluoropyrimidine-related toxicity, including for example stomatitis, diarrhoea, mucosal inflammation, neutropenia and neurotoxicity. Treatment with 5-Fluorouracil, capecitabine or tegafur-containing medicinal products is contraindicated in patients with known complete DPD deficiency. Consider a reduced starting dose in patients with identified partial DPD deficiency. Initial dose reduction may impact the efficacy of treatment. In the absence of serious toxicity, subsequent doses may be increased with careful monitoring. Therapeutic drug monitoring (TDM) of 5-Fluorouracil may improve clinical outcomes in patients receiving continuous 5-Fluorouracil infusions.
- **Hand-foot syndrome (HFS),** also known as palmar-plantar erythrodysaesthesia (PPE) has been reported as an unusual complication of high dose bolus or protracted continuous therapy for 5-Fluorouracil (see table 3 for dose modifications).

DRUG INTERACTIONS:

- Current SmPC and drug interaction databases should be consulted for information.

REFERENCES:

1. Kuebler JP, Wie and HS et al. Oxaliplatin combined with weekly bolus fluorouracil and leucovorin as surgical adjuvant chemotherapy for stage II and III colon cancer: results from NSABP C-07. J Clin Oncol. 2007 Jun 1;25(16):2198-204.
2. Giraud E L, Lijster B D, et al. Dose recommendations for anticancer drugs in patients with renal or hepatic impairment: an update. Available at: https://www.thelancet.com/journals/lanonc/article/PIIS1470-2045(23)00216-4/fulltext
3. Dosage Adjustment for Cytotoxics in Hepatic Impairment January 2009; North London Cancer

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Network.
4. NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting. V6 2025 Available at: https://www.hse.ie/eng/services/list/5/cancer/profinfo/chemoprotocols/nccp%20antiemetic%20classification%20document%20v1%202018.pdf
5. HPRA Direct Healthcare Professional Communication. 5-Fluorouracil (i.v.), capecitabine and tegafur containing products: Pre-treatment testing to identify DPD-deficient patients at increased risk of severe toxicity. Accessed Aug 2020 Available at: https://www.hpra.ie/docs/default-source/default-document-library/important-safety-information-from-marketing-authorisation-holders-of-products-containing-5-fluorouracil-(i-v-)-capecitabine-and-tegafur-as-approved-by-the-hpra.pdf?sfvrsn=0
6. Oxaliplatin 5mg/mL Summary of Product Characteristics. Accessed May 2025 .Available at: https://assets.hpra.ie/products/Human/27562/Licence_PA2315-114-001_02052025141313.pdf
7. Fluorouracil 25mg/mL infusion for injection. Accessed May 2025 . Available at: https://www.medicines.ie/medicines/fluorouracil-25-mg-ml-solution-for-injection-or-infusion-32184/patient-info

Version	Date	Amendment	Approved By
1			Prof Maccon Keane
2	01/05/19	Updated treatment table for folinic acid	Prof Maccon Keane
3	09/10/2019	Updated tests, exclusions, drug interactions.	Prof Maccon Keane
4	07/01/2020	Standardisation of treatment table. Updated exclusion criteria. Updated recommended dose modifications for oxaliplatin in renal impairment.	Prof Maccon Keane
5	26/02/2020	Standardisation of treatment table.	Prof Maccon Keane
6	27/05/2020	Regimen reviewed.	Prof Maccon Keane
7	02/09/2020	Updated exclusion criteria, baseline testing, dose modifications and adverse events with respect to DPD deficiency as per DHPC from HPRA June 2020. Updated Adverse events regarding palmar-plantar erythrodysaesthesia. Treatment table updated regarding 5-Fluorouracil bolus administration.	Prof Maccon Keane
7a	21/11/2023	Formatting changes and grammatical corrections.	NCCP
7b	03/03/2025	Additional wording added to baseline test section.	NCCP
8	14/07/2025	Regimen reviewed. Updated ICD-10 codes . Updated exclusions section. Addition of cautions section.Renal and hepatic dose modifications table aligned to Giraud et al 2023. Regimen updated in line with NCCP standardisation.	Prof Maccon Keane

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